GEOMETRY PROPOSES, JUDGEMENT DISPOSES: ONLINE PROPOSAL AGGREGATION FOR DELIBERATIVE AI

Anonymous authors
Paper under double-blind review

ABSTRACT

Scaling participatory collective decision-making requires aggregating high volumes of free-form citizen proposals in real time. Standard natural language processing tools face a fundamental dilemma: off-the-shelf sentence embeddings reliably capture topical relevance but fail catastrophically on stance-critical distinctions, while unassisted large language models (LLMs) incur prohibitive quadratic token costs, context limits, and loss of reference precision over long candidate lists. We investigate a streaming architecture built on the empirical doctrine: *geometry proposes, judgement disposes*. Cosine similarity over dense text embeddings is demoted strictly to approximate-nearest-neighbor candidate generation, while all state-mutating placements are decided by discrete, fail-closed LLM adjudicators operating under asymmetric doubt. Furthermore, an opt-in Claim Registry reframes continuous deduplication into streaming classification across an evolving codebook of canonical claims anchored by verbatim exemplars. Evaluated on the 875 hard counterfactual triplets from Blair et al. (2026b), discrete judged placement over raw anchors achieves 95.0%–96.5% accuracy, compared to 0.5% for raw cosine and 80.0% for decorrelated preference tuning, whereas unconstrained classification against unconsolidated 94-claim codebooks drops to 36.0%–55.8% on strict own-claim scoring, motivating geometric candidate pruning. On an adversarial 100-statement streaming benchmark, the synthesis layer achieves exact pair recovery (150/150 ground-truth pairs merged across three arrival orders, Adjusted Rand Index $= 1.000$; composite score 0.884–0.910 in English, ≈ 0.930 in Hebrew). Finally, we report an end-to-end replay on 114 authentic Hebrew statements from environmental researchers. Independent cross-model-family blind audits measured an initial member-level merge precision of 65% and text fidelity of 0.647 due to multi-hop snowballing and silent misses. Incorporating a transitivity check on merge sweeps, a revisit pass, and explicit writer fidelity clauses improved merge precision to 74.4% (with no cluster exceeding 4 members) and fidelity to 0.953 with zero lost participant voices. We report this 74% precision, rather than idealized benchmark numbers, as the realistic production expectation on uncurated corpora.

1 INTRODUCTION

Democratic institutions, municipal assemblies, and digital citizen platforms (such as Polis and Remesh) increasingly invite participants to submit free-form policy proposals and evaluate statements authored by peers. While expressive natural language enables nuanced public input, it inevitably produces severe corpus fragmentation: when dozens of participants submit the same policy intervention in distinct wordings, an unaggregated corpus fragments collective support. Fifty citizens proposing the exact same municipal reform appear to a consensus engine as fifty disparate minority factions.

To overcome fragmentation without compromising democratic legitimacy, an online aggregation engine must fulfill three continuous operational mandates:

1. **Synthesis:** Detecting when distinct submissions express propositionally equivalent policy proposals and merging them into a unified statement that pools peer evaluations.

2. **Theming:** Grouping distinct proposals under topical headings to structure public navigation, without asserting that constituent proposals agree on specific interventions.
3. **Standing Ideas:** Preserving unique submissions as sovereign, visible public proposals immediately from the moment of their creation.

These operations are governed by three systems requirements: (1) **Streaming Execution:** Proposals arrive asynchronously and must be placed incrementally without waiting for batch re-clustering; (2) **Order Independence:** Decision-critical synthesis should not depend arbitrarily on arrival sequence; and (3) **Multilingual Robustness:** The pipeline must support morphologically rich languages (such as Hebrew alongside English).

Crucially, two fundamental distinctions must be maintained across all system components:

$$\text{Same Topic} \neq \text{Same Proposal} \qquad \text{and} \qquad \text{Same Wording} \neq \text{Same Stance} \tag{1}$$

The first distinction separates thematic proximity from propositional equivalence: "install speed bumps near schools" and "lower speed limits near schools" share a topic (traffic safety) but represent distinct municipal interventions involving different budgets and enforcement agencies. Merging them misrepresents both authors and distorts democratic consent. The second distinction separates surface lexical overlap from normative alignment: "ban private vehicular traffic downtown" and "do not ban private vehicular traffic downtown" share nearly identical vocabulary and syntax, yet their policy stances are diametrically opposed.

In this work, we demonstrate that off-the-shelf sentence embeddings collapse on stance-decoupled proposals, while naive LLM architectures incur severe scaling and list-length bottlenecks. We introduce an end-to-end deliberative aggregation architecture governed by the doctrine: *geometry proposes, judgement disposes*.

2 Related Work

Deliberation Systems and Opinion Mapping. Platforms such as Polis (Small et al., 2021) map opinion spaces by applying PCA to sparse vote matrices. Generative social choice (Fish et al., 2026; Boehmer et al., 2025) produces representative statement slates via LLM summarization, while computational social choice formalizes proportionally fair clustering (Chen et al., 2019; Kellerhals & Peters, 2024) and facility location in metric spaces (Feldman et al., 2016). Complementary work investigates consensus learning theory (Blair et al., 2026a), verbal reward design (Blair et al., 2025), and deliberative alignment (Tessler et al., 2024).

Sentence Encoders and Preference Geometry. Standard sentence encoders (Reimers & Gurevych, 2019; Muennighoff et al., 2023) optimize semantic textual similarity, web retrieval, or natural language inference. Contrastive learning methods like SimCSE (Gao et al., 2021) pull entailments together and push contradictions apart. Following the representation invariance formulation of Achille & Soatto (2018), Blair et al. (2026b) formalize the preference-semantics mismatch as an invariance problem: an embedding $\psi(x) \in \mathbb{R}^d$ decomposes into an in-subspace preference component $\psi_S(x) \in S$ (dimension $k \ll d$) and an orthogonal semantic nuisance $\psi_\perp(x) \in S^\perp$ (capturing topic, style, and vocabulary):

$$\psi(x) = \psi_S(x) + \psi_\perp(x), \quad \text{with } \langle \psi(a), \psi(x) \rangle = \langle \psi_S(a), \psi_S(x) \rangle + \langle \psi_\perp(a), \psi_\perp(x) \rangle \tag{2}$$

Standard cosine weights both components equally. While stance and vocabulary correlate on natural corpora, when stance and wording decouple on hard counterfactual triplets, the nuisance margin Δ_T overwhelms the preference margin Δ_S. Blair et al. (2026b) address this by repairing the geometry via Decorrelated Preference Tuning (DPT, achieving 80.0% accuracy) or per-topic vote projections (81.1%). In contrast, our approach demotes the geometry, treating embeddings purely as approximate-nearest-neighbor candidate generators for discrete, fail-closed adjudication.

LLMs as Evaluators, Judges, and Clusterers. The use of LLMs as zero-shot evaluators and judges has expanded across NLP (Zheng et al., 2023). While instruction-tuned models exhibit strong pairwise discriminative capabilities, prior work highlights vulnerabilities including prompt order sensitivity, position bias, and extreme prompt variability (Wang et al., 2023; Sclar et al., 2024; Liu et al., 2024). In parallel, LLM-guided clustering has been explored for discovery and taxonomy construction (Zhang et al., 2023; Viswanathan et al., 2024). Our work connects to retrieve-then-rerank pipelines (Nogueira & Cho, 2019; Lewis et al., 2020), but differs in two key respects: our reranker returns a typed relational discrete verdict rather than a continuous scalar score, and classifies against an evolving, dynamically consolidated codebook rather than a static document corpus.

3 Why Neither Metric Embeddings Nor Pure LLMs Suffice

3.1 The Stance-Blindness of Metric Embeddings

Direct cosine thresholding over dense vector spaces exhibits near-zero decision-grade precision on stance-decoupled proposals. On the 875 hard counterfactual triplets of Blair et al. (2026b), 100.0% of stance-flipped distractors cleared the 0.60 topical clustering threshold and 99.1% cleared the 0.85 merge threshold, yielding an end-to-end triplet accuracy of exactly 0.5% on `text-embedding-3-small`. A threshold that admits over 99% of counterfeits has no discriminative power.

However, nearest-neighbor vector search over the same embedding space recovered the true paraphrase partner among the top candidates for 100/100 English statements under `text-embedding-3-small`. In Hebrew, separability under 3-small is inadequate (recovering the true partner as nearest neighbour in only 56/100 statements); upgrading to `text-embedding-3-large` (truncated natively to 1536 dimensions) lifts Hebrew top-1 recovery to 88–89/100. Thus, vector geometry retains strong utility for candidate retrieval, but cannot serve as a decision rule.

3.2 Systems Bottlenecks and List-Scale Degradation of Unassisted LLMs

Conversely, deploying an unassisted LLM to cluster streaming proposals directly without geometric filtering encounters severe practical limits:

1. **Loss of Reference Precision at List Scale:** When prompted to classify an incoming proposal against a growing, unpruned list of claims, the judge's ability to identify the *specific* matching claim degrades sharply. As established in Section 5.1, the same judge that scores 95.0% in focused candidate comparison drops to 36.0%–55.8% under strict own-claim scoring against codebooks of ≈ 94 claims. The degradation is one of reference rather than of stance discrimination: match-to-any-equivalent-claim remains at 74.4%–92.9%, and the distractor false-attach rate does not rise with list length (10.3% at 94 claims versus 12.8% at a single claim). Long candidate lists nonetheless carry documented costs in position bias and context-length degradation (Liu et al., 2024), and prompt size grows with the codebook, which together motivate geometric pre-filtering and periodic codebook consolidation.
2. **Linear Prompt Growth and Quadratic Token Cost:** In an offline batch baseline where an LLM clustered all 100 benchmark statements simultaneously in one prompt, the model produced the identical ground-truth synthesis partition across three arrival orders (150/150 pairs recovered, F1 $= 1.000$) and created cleaner topical headings. Furthermore, an incremental bare-LLM baseline (placing statements incrementally without vector gating) achieved a composite score of 0.937–0.984 (synthesis F1 1.000, topic F1 0.844–0.960 across 101 calls per run at $\approx$ \$0.03). However, in streaming deployments, the prompt size of an unassisted LLM grows linearly with the number of active proposals, resulting in cumulative $O(N^2)$ token ingestion. At civic deliberation scale (10^3 to 10^4 statements), prompts exhaust context budgets and enter the list-scale regime above.
3. **Scope Inflation and Lack of Fail-Closed Custody:** Generative models drift toward helpful generalization (e.g., drifting "subsidized transit for seniors" into "better transit for all"), altering civic commitments. Furthermore, unassisted LLMs lack fail-closed state custody during API timeouts or malformed JSON responses.

4 Pipeline Architecture: Geometry Proposes, Judgement Disposes

The online placement architecture executes placement through a strictly ordered, seven-pass cascade designed for real-time incremental execution.

4.1 Representation Layer: Context Contracts and Brief Distillation

To prevent boilerplate framing from compressing vectors into an artificial ≈ 0.75 floor, statements ≥ 40 characters are distilled by an ultra-fast worker LLM into an objective 5–15 word policy brief. Embeddings are generated from the brief under a fixed context contract:

$$\text{Input} = \text{“Question: ”} + \langle\text{Deliberation Question}\rangle + \text{“\textbackslash nAnswer: ”} + \langle\text{Distilled Brief}\rangle \tag{3}$$

The original verbatim statement is preserved untouched for public display and discrete adjudication.

4.2 Calibrated Bands and Multi-Stage Member Evidence

Candidate discovery utilizes approximate-nearest-neighbor (ANN) search (top $k = 15$) over a partitioned vector index. Cosine similarity is partitioned into calibrated bands: `attachThreshold` (0.85 for 3-small, 0.86 for 3-large), `synthLowerBound` (0.78 / 0.80), `clusterThreshold` (0.60 / 0.75), and `reviewLowerBound` (0.45 for both 3-small and 3-large, serving as the ANN search floor). Crossing a band makes an action eligible; it never triggers the action.

Evidence for existing syntheses is evaluated via multi-stage member expansion: the arithmetic mean of the top-2 constituent member similarities:

$$\text{Score}_{\text{cluster}}(s) = \frac{\cos(\psi(s), \psi(m^{(1)})) + \cos(\psi(s), \psi(m^{(2)}))}{2} \tag{4}$$

Requiring two independent members to agree prevents an isolated outlier from dragging a broad cluster across a merge threshold.

To prevent snowball collapse, receiving clusters must satisfy conjoint cohesion gates before reaching an LLM judge: (1) **Centroid Proximity:** Cosine similarity between candidate and cluster centroid must exceed $\tau_{\text{cohesion}} = 0.815$ (empirically calibrated: median genuine attach is 0.898 vs. 0.723 for same-topic non-matches); and (2) **Quorum:** At least 50% of cluster members must individually exhibit cosine ≥ 0.78 to the candidate.

4.3 The Seven-Pass Placement Cascade

Arrivals traverse a strict priority hierarchy:

- **Pass 1 (Synthesis Attach — P1):** Merges into an existing synthesis if top-2 evidence ≥ 0.85, cohesion passes, and discrete LLM equivalence verdict is `same`.
- **Pass 2 (Synthesis Spawn — P2):** If P1 fails, tests whether the statement can be paired with an unattached statement to spawn a new synthesis (pair cosine ≥ 0.78, LLM coherence verified).
- **Pass 3 (Judged Theme Filing — P3):** If no synthesis is warranted, an LLM files the statement into an existing theme judged against member contents (doubt $\rightarrow$ `NONE`).
- **Pass 4 (Claim Registry — P4):** Classifies the proposal against an evolving canonical codebook.
- **Pass 5 (Raw Theme Spawn — P5):** Disabled by default; raw-pair theme creation caused black-hole collapses in early testing. Instead, themes are born from syntheses: whenever a newly spawned synthesis fits no existing theme, `createThemeForSynth` creates a dedicated theme for it.
- **Pass 6 (Review Queue — P6):** Borderline candidates ($0.45 \leq \cos < 0.60$) logged for human review.
- **Pass 7 (Standing Singleton — P7):** Independent standing idea visible immediately.

The Priority Invariant: Synthesis (P1/P2) strictly precedes Theming (P3). Theming membership is non-terminal, while synthesis membership is terminal and enforced by authoritative reverse-index lookups.

4.4 Discrete Equivalence Adjudication under Asymmetric Doubt

The P1 judge classifies raw text pairs into $V \in \{\texttt{same}, \texttt{related}, \texttt{different}, \texttt{opposite}\}$ under the operational rubric:

> “`same`: A and B are paraphrases of essentially the same proposal. `related`: same topic but different actions, stances, or magnitudes. `different`: about different things; wording

> similar by coincidence. `opposite`: contradictory actions on the same subject ('ban X' vs 'permit X'). Use 'same' ONLY when the two proposals are essentially interchangeable. When in doubt between 'same' and 'related', prefer 'related'."

Only `same` authorizes a merge. Under asymmetric doubt, prompt rubrics systematically bias errors toward the lower-cost outcome (delaying a merge to sweeps rather than executing a false merge).

Verdicts are stored in a persistent content-addressable cache keyed by an order-independent SHA-1 hash of the normalized statement pair, versioned by model and prompt version. Fallback verdicts resulting from API timeouts or errors are never cached. Fault tolerance enforces:

$$\text{Placement Path: Fail-Closed (Error} \rightarrow \texttt{different}) \quad \text{and} \quad \text{Custody Path: Fail-Open (Error} \rightarrow \text{Keep)} \tag{5}$$

4.5 The Claim Registry: Dynamic Codebook Construction

The Claim Registry addresses a structural blind spot of the placement cascade: the cascade can only evaluate candidate pairs that cosine surfaced, leaving a semantic recall gap for distant paraphrases that no threshold tuning can close. The registry reframes deduplication into streaming classification over an evolving codebook of canonical claims. Each claim $C_k = \langle T_{\text{canonical}}, E_{\text{public}}, X_{\text{anchor}} \rangle$ contains a 5–15 word canonical wording, a public explanation, and the unedited verbatim anchor exemplar. The registry is opt-in per question and ships default-off.

Within the registry, cosine survives in its demoted role as a *ranker*: it orders the codebook most-plausible-first so that likely matches appear early in the prompt, mitigating the position bias documented by Liu et al. (2024), and it filters nothing.

The registry maintains stability via four protocols:

1. **Fixed-Point Consolidation:** Every 15 placements, an LLM merges redundant claims under the rubric: "same-topic-different-action is NOT a merge… no merges is a perfectly good answer" (claims graduate to `confirmed` only after $\geq$ 3 members and surviving a pass untouched). Because every consolidation merge represents an observed false-negative (a 'new' verdict) during online placement, the running consolidation merge count provides an autonomous online estimate of the registry's own recall error.
2. **Broaden-Ratchet:** Rewordings are classified into $\{\texttt{reword}, \texttt{broaden}, \texttt{narrow}, \texttt{different}\}$. Because individually harmless broadenings compose, each claim retains its founding anchor X_{anchor}, and after two consecutive broadenings the new wording is tested directly against that anchor.
3. **Hierarchical Routing with Ungated Fallback:** Above 30 claims, classification routes through at most two topical claim groups, but any `NONE` verdict immediately triggers an ungated flat fallback. In corpus replays, hard partitioning without fallback caused 59% of all recall errors, whereas the ungated fallback lifted corpus-replay placement accuracy from 61.3% to 75.3% ($p = 7.5 \times 10^{-4}$).
4. **Typed Opposition:** When a proposal negates a claim, the classifier emits `opposes`, creating a typed symbolic counter-edge $\langle s_i, \text{OPPOSES}, C_k \rangle$.

4.6 Synthesis Generation and Safety Invariants

When statements merge, synthesis text generation enforces strict invariants: (1) **Dual Refusal:** The writer refuses synthesis on directional conflict or distinct actions within the same theme ("if a supporter could reasonably back one while opposing the other's intervention, the inputs are distinct ideas; when in doubt, refuse"); (2) **The Scope Rule:** An explicit never-widen-scope rule mandates that synthesized texts cannot generalize specific populations (in offline A/B testing over the 50 ground-truth pairs, this rule eliminated scope inflation from 4/50 to 0/50, while live benchmark audits confirmed 98/98 member intents preserved); (3) **The Formulation Clause:** "formulation differences are not grounds for refusal; refuse only when the difference IS the intervention" (elevating Hebrew twin recovery from 45/50 to 49/50 offline); (4) **Evaluator-Share Weighting:** Input framing is weighted by accumulated peer evaluations; and (5) **Unicode Verification:** Output language is enforced by detecting the Unicode script range of inputs. Unconstrained, 8 of 10 topic titles for an all-English corpus were emitted in German or Spanish; because cluster titles are embedded and reused as candidate retrieval evidence, wrong-language titles directly corrupt vector retrieval rather than merely creating a display flaw.

Table 1: End-to-end accuracy on the 875 hard counterfactual triplets (Blair et al., 2026b).

Model / Configuration	Decision Mechanism	Triplet Accuracy (%)
Raw `text-embedding-3-small` (Pipeline Gate)	Metric Cosine	0.5%
Production Context Format (3-small)	Metric Cosine	1.8%
Base Sentence-T5-XL (Blair et al., 2026b)	Metric Cosine	48.3%
e5-large-v2 / BGE-large-en-v1.5	Metric Cosine	26.7% / 6.3%
DPT-Tuned Sentence-T5-XL (Blair et al., 2026b)	Repaired Metric	80.0%
Rank-20 Per-Topic Projection (Blair et al., 2026b)	Vote-Trained Metric	81.1%
Unconsolidated Codebook ($\approx$ 94 claims), Strict Own-Claim	Discrete LLM Classification	36.0%–55.8%
Unconsolidated Codebook ($\approx$ 94 claims), Any Equivalent Claim	Discrete LLM Classification	74.4%–92.9%
Registry: Generated Canonicals Alone	Discrete LLM Classification	70.1%
Registry: Enriched + Stance Caution	Discrete LLM Classification	88.6%
Registry: Raw-Anchor Claims (Historical, `gpt-4o-mini`)	Discrete LLM Classification	95.0%
Registry: Raw-Anchor Claims (Stronger Model, `gpt-4o`)	Discrete LLM Classification	96.5%

5 Empirical Evaluation

5.1 Hard-Triplet Benchmark (Shared Instrument with Blair et al., 2026)

We evaluate on the 875 hard counterfactual triplets released by Blair et al. (2026b) across 11 deliberation datasets from Polis, Remesh, and Generative Social Choice. Scoring is end-to-end: a triplet is correct if and only if the preference match attaches to the anchor claim and the distractor is rejected.

As shown in Table 1, off-the-shelf cosine collapses (0.5%–48.3%). While DPT (80.0%) and per-topic projection (81.1%) significantly improve metric scoring, discrete judged placement over raw anchors achieves 95.0%–96.5% ($p \approx 10^{-249}$ vs. raw cosine under exact McNemar testing).

Four caveats must be emphasized:

1. **Canonicalization Compression Loss:** Moving from raw-anchor representations (95.0%) to bare generated canonicals (70.1%) incurs a 24.9 percentage-point drop. Enriched claims (adding public explanations and verbatim exemplars) recover accuracy to 88.6%, showing that compression loss in canonicalization is the primary vulnerability.
2. **Embedding-Gate Equivalence:** A standard RAG baseline (embedding retrieval followed by LLM judge) also achieves 95.0% on these English triplets. Thus, the registry's marginal value is not raw triplet accuracy on clean English rewrites, but cold-start robustness and cross-lingual recall.
3. **What the Codebook-Scale Numbers Do and Do Not Show:** Evaluating the same judge against an unpruned list of $\approx$ 94 claims drops strict own-claim accuracy to 36.0%–55.8%, but the decomposition matters. The benchmark codebook is deliberately unconsolidated and assigns every anchor its own claim, so it contains many word-for-word equivalent entries; roughly a third of matches attach to one of those legitimate duplicates rather than to the anchor's own claim. Match-to-any-equivalent-claim accordingly remains at 74.4%–92.9%, and distractor false-attach does not rise with list length. The operational conclusion is therefore narrower than the headline figure suggests: list scale degrades *reference precision* and inflates prompt cost, which retrieval gating and periodic consolidation address, but it does not collapse the judge's stance discrimination.
4. **Distractor Construction Limits:** The evaluated distractors were constructed specifically to defeat sentence embeddings (maximizing lexical overlap while flipping stance). Adversarial distractors crafted specifically against language models (incorporating subtle hedges, buried negations, or presupposition shifts) would present a harder challenge; thus 95.0%–96.5% should be interpreted as an upper bound.

5.2 Model Upgrades and Pilot-150 Regression Analysis

Replicating the benchmark on a fixed, dataset-stratified 150-triplet subsample with the upgraded reasoning model (`gpt-5.6-luna`) revealed key operational insights:

- **Raw-Anchor Stability:** Raw-anchor accuracy was robust across model generations (98.7% under `gpt-4o-mini` vs. 96.7% under `gpt-5.6-luna`, $p = 0.45$).
- **The Enriched-Condition Regression:** The enriched condition regressed significantly from 92.0% to 79.7% ($p = 2.8 \times 10^{-4}$), falling to statistical parity with DPT. Inspection revealed that the upgraded model's generated canonicals dropped subtle stance qualifiers (e.g., an illegal-with-life-exception anchor canonicalized as "abortion is permitted when pregnancy poses a genuine risk to the mother's life"), after which the classifier correctly matched the stance-flipped distractor to the stance-lossy claim. A prompt fix demanding explicit stance conditions restored accuracy partially to 82.7%, confirming that compressive canonicalization is an inherent source of loss.
- **Reasoning Token Headroom:** The legacy 300-token completion budget caused reasoning models to exhaust tokens internally before emitting JSON, triggering fail-closed aborts on 2 of 150 triplets. Reserving a 2,000-token reasoning headroom resolved all deterministic aborts, lifting raw-anchor accuracy to 98.0%.

5.3 Live Streaming Benchmark: Orthogonal Layer Variance

We evaluated the full streaming pipeline on an adversarial corpus of 100 statements (10 topics $\times$ 5 idea-groups $\times$ 2 near-paraphrases; 50 ground-truth syntheses) streamed in random order across three seeds (42, 7, 1234):

- **Synthesis Layer:** Across all three arrival orders, the synthesis layer achieved exact, order-independent recovery: 150/150 ground-truth pairs merged with zero false merges across 450 placements (Precision $= 1.000$, Recall $= 1.000$, Adjusted Rand Index $= 1.000$).
- **Theming Layer:** In contrast, theming remained order-sensitive (topic F1 0.711–0.775, generating 10–17 headings). The final composite accuracy reached 0.884–0.910 in English and ≈ 0.930 in Hebrew (synthesis F1 0.947, 45/50 pairs).
- **Cross-Generator Control:** To verify that benchmark performance is not an artifact of prompt-corpus co-design, an independent vendor's model authored a second adversarial corpus with compressed geometry, where the best cosine-only F1 was capped at 0.871. Our judged pipeline achieved a synthesis F1 of 0.980 with zero false merges, exceeding the metric ceiling by 10.9 percentage points.

5.4 Authentic Civic Deliberation: A Case Study on Real Deliberation Data

To evaluate behavior on real, uncurated deliberation data, we replayed an authentic question from a deliberation convened by a sustainability research institute, involving 114 Hebrew statements from environmental researchers on research translation. Statements were ordered chronologically and replayed through a local production emulator under `text-embedding-3-large`. The Claim Registry (§4.5) was *disabled* in both replays, as it is by default; the near-verbatim duplicates identified by the audit below therefore constitute independent motivation for it rather than evidence against it.

Cross-Model-Family Blind Audit. To evaluate outputs without an objective answer key, two independent auditor models and two missed-merge sweeper models from an alternate foundation model family evaluated the results blind to titles, descriptions, and each other under refutation rubrics. Only findings agreed upon by both independent auditors were counted. An automated text fidelity scorer evaluated whether published synthesis texts preserved constituent member asks.

Audit Findings and Engineering Fixes. As detailed in Table 2, the first judged replay significantly improved structure over legacy production (which had created an uninterpretable 24-member blob). However, the blind audit revealed that clean benchmark numbers did not transfer directly: merge precision was 65%, 19 missed groupings were silent (only 2 of 21 missed statements were logged in the review queue), and fidelity was 0.647 with 3 lost voices due to snowballing in an 8-member cluster.

Three targeted algorithmic fixes were engineered: (1) **Transitivity Check on Merge Sweeps:** Merging two syntheses requires the two most distant cross-cluster members to be explicitly judged as `same`, inspecting up to 6 members per side; (2) **Scheduled Revisit Pass:** Un-synthesized statements whose similarity evidence clears the synthesis band during sweeps are reinjected into the full judged

Table 2: Deliberation structure and blind audit metrics across replays on 114 authentic Hebrew statements.

Metric / Operational Axis	Legacy Production	First Judged Replay	Fixed Build Replay
Syntheses Count	1 (holding 24 stmts)	11 (holding 34 stmts)	20 (holding 43 stmts)
Largest Synthesis Size	24 members	8 members (snowball)	4 members
Themes Count	3 (one flat theme of 59)	2 themes	3 themes
Statements in No Cluster	14 (silently unplaced)	21	22
Review-Queue Events Logged	Not Measured	30	42
Member-Level Merge Precision	Not Measured	64.7% (22/34)	74.4% (32/43)
Clean Syntheses (0 refutations)	Not Measured	6 of 11	11 of 20
Agreed Missed Merges	Not Measured	11 attach + 5 spawn (19 silent)	12 attach + 3 spawn (0 silent)
Text Fidelity Score	Not Measured	0.647	0.953
Lost Member Asks	Not Measured	3 lost, 9 weakened	0 lost, 2 weakened
Fabricated Commitments	Not Measured	2 invented mechanisms	0 (1 soft benefit)

Table 3: Architectural comparison: geometry repair vs. geometric demotion.

Dimension	Geometry Repair (Blair et al., 2026b)	Geometric Demotion (Ours)
Primary Decision Rule	Linear Scorer in Mapped Space	Discrete LLM Adjudication
Role of Vector Space	Final Arbiter of Placement	Candidate Generator (ANN, $k = 15$)
Hard-Triplet Accuracy	80.0% (DPT) / 81.1% (Projected)	95.0%–96.5%
Per-Placement Cost	$O(1)$ Vector Operations	1–2 Discrete LLM Calls
Target Relation	Continuous Preferential Utility	Propositional Equivalence
Representation of Opposition	Metric Distance in Subspace S	Symbolic Typed Counter-Edge

cascade; and (3) **Writer Invariants:** Strict clauses forbid adding unstated mechanisms and mandate that every input's specific ask remains recognizable.

On the fixed build replay, member-level precision rose to 74.4% (32/43 correct; all errors confined to small pairs ≤ 4 members), text fidelity rose to 0.953 with zero lost voices, and structurally silent misses dropped to zero (all 15 agreed misses were revisit-refused or awaiting sweep rounds). We report this 74% precision as the realistic production expectation on authentic, highly correlated civic corpora.

6 Discussion: Repairing vs. Demoting Geometry

Divergent Relational Targets. As summarized in Table 3, the two paradigms address distinct relations. Blair et al. (2026b) model preferential similarity (whether participant v endorses statement j), a continuous quantity suited for ideal-point metric spaces. We model propositional equivalence (whether two statements constitute the same policy intervention), a discrete, symmetric equivalence relation requiring discrete adjudication. Furthermore, metric distances cannot represent direct political opposition; our architecture represents opposition as an explicit typed graph edge.

The Natural Hybrid. Because our architecture confines geometry to candidate generation, Blair et al.'s learned preference projections integrate naturally as an upgraded candidate generator: Candidate Pool $= \mathrm{ANN}_{\mathrm{Projected}\ L^\top\psi}(s, k = 15) \rightarrow$ Discrete Judge. This hybrid promises to close cross-lingual recall gaps and reduce judge invocation by separating overlapping distributions, while providing downstream social choice with verified, deduplicated proposals.

Design Implication for Theming. Because themes merely organize public browsing without altering what participants consented to (unlike syntheses, which pool voting power), themes can be periodically re-synthesized in batch from current syntheses in a single inexpensive call. This architectural separation explains why batch baselines outperform streaming heuristics on theming, providing a principled design recommendation for deliberative platforms.

7 LIMITATIONS AND OPEN PROBLEMS

- **Single-Vendor Foundation Model Dependency:** Despite modular codebase interfaces (including files named `gemini.ts`), every production LLM component (equivalence judge, synthesis writer, claim classifier) currently runs on OpenAI models (`gpt-4o-mini`, `gpt-4o`, `gpt-5.6-luna`). While 5% shadow audits evaluate drift, single-vendor cognitive bias remains an active limitation.
- **Synthetic vs. Natural Distribution Shifts:** The 875 triplets are counterfactual rewrites. Blair et al.'s natural-data accuracies (65%–69%) indicate that most natural proposal pairs are topically and lexically separable, while the hard tail is what our architecture is designed to handle.
- **Benchmark Codebook Consolidation:** The codebook-scale conditions of Table 1 were evaluated on an unconsolidated codebook, which penalizes attachment to equivalent duplicate claims. A consolidated-codebook replication would isolate genuine list-scale confusion from this scoring artifact and has not yet been run.
- **Residual Error Modes on Real Data:** In the real-data deliberation replay, precision remains bounded at 74.4%. Residual errors stem from the spawn writer accepting related Hebrew pairs as `same`, and revisit sweeps exhausting round budgets.
- **Theming Stochasticity:** While synthesis achieves order-independent convergence (ARI $= 1.000$), online thematic clustering remains order-sensitive (F1 0.711–0.775). Developing a principled online theming objective is an open problem.
- **Inference Cost and Latency:** Production per-placement latency averages 1–2 seconds (8–10 seconds when the writer is invoked), requiring 2.85 LLM calls per statement at an estimated cost of \$0.20–\$0.50 per 1,000 statements, with verdict-cache hits resolving in ≈ 45 ms. In contrast, pure vector operations execute in sub-millisecond time.

8 CONCLUSION

Aggregating free-form civic proposals requires a disciplined separation of concerns: geometry proposes, judgement disposes. Dense vector spaces provide fast candidate discovery, while calibrated, fail-closed LLM adjudicators enforce propositional equivalence under asymmetric doubt. Validated across 875 hard triplets, live streaming benchmarks, and an authentic deliberation case study, this architecture provides an empirically grounded foundation for AI-assisted collective decision-making.

REPRODUCIBILITY STATEMENT

Complete production pipeline implementations, Claim Registry modules, benchmark evaluation test harnesses, Pilot-150 replication scripts, and blind audit logs are included in the supplementary material and will be made publicly available in an open-source repository upon publication. Detailed prompt rubrics, parameter grids, and algorithmic flows are provided in Appendices A–C.

ETHICS AND BROADER IMPACT STATEMENT

Deliberative aggregation systems directly shape public consensus and policy recommendations. Erroneous merges or unauthorized scope inflation distort democratic representation. Our architecture mitigates these risks via fail-closed error handling, asymmetric doubt, the non-expansion scope rule, and typed opposition edges. Real deliberation data was exported read-only and without personal participant identifiers, and no participant statement is quoted in this paper.

AI USE STATEMENT (ICLR 2027 COMPLIANCE)

In accordance with the ICLR 2027 Artificial Intelligence Policy, the authors disclose the following details regarding AI tool usage.

Required Disclosures.

1. **Core System Implementation and Execution.** Large language models are foundational operational components of the deliberative aggregation pipeline:

- *Brief Distillation:* A lightweight model (historical evaluations used `gpt-4o-mini`; current production uses `gpt-5.6-luna`) extracts 5–15 word briefs from proposals ≥ 40 characters prior to vector embedding.
- *Discrete Equivalence Adjudication:* Instruction-tuned models (`gpt-4o-mini`, `gpt-4o`, `gpt-5.6-luna`) execute 4-way classification (`same`, `related`, `different`, `opposite`) under asymmetric doubt rubrics.
- *Thematic Filing and Claim Classification:* Models classify proposals against theme contents and dynamic codebooks.
- *Synthesis Text Generation:* Generative language models author consensus proposals under explicit refusal constraints, evaluator-share weighting, and scope invariants.

2. **Synthetic Data Generation.** Synthetic benchmark data was employed to establish controlled ground-truth conditions:
   - The 875 hard counterfactual evaluation triplets from Blair et al. (2026b) were generated using GPT-4o to rewrite human deliberation anchors into preference matches and stance distractors.
   - The 100-statement live streaming corpus was synthetically constructed across 10 topics, 5 idea groups, and 2 paraphrases per group. A second, independent corpus used in the cross-generator control was authored by a different vendor's model.
3. **Multilingual Processing.** LLM distillation routines and Unicode script verification were applied to process Hebrew and English proposals.
4. **Qualitative Auditing and Blind Evaluation.** In the real-data case study (where no answer key exists), two independent auditor models and two missed-merge sweepers from an alternate foundation model family were used to conduct cross-model blind audits.

Recommended Disclosures.

1. **Drafting and Editorial Assistance.** Google Gemini was employed for drafting assistance, structural condensation to conform to the ICLR 9-page limit, and LaTeX typesetting. A second model was used to audit the resulting manuscript against the underlying source code and run artefacts.
2. **Literature Search.** LLMs assisted in surveying relevant literature across computational social choice and LLM evaluation.

Human Verification and Scientific Responsibility.

- All mathematical formulations, complexity analyses, and empirical calibrations (such as the 0.815 cohesion floor calibrated over 4,900 candidate pairs) were verified manually.
- All empirical metrics, benchmark results, and audit logs trace directly to certified execution runs and logs in the project repository.
- Language-model-drafted reference entries were found to contain fabricated titles and author lists in earlier revisions. Every entry in the present bibliography was therefore re-derived by the human authors from the ACL Anthology, the official conference proceedings, or the arXiv record, and no reference entry in this manuscript was authored by a language model.

The authors take full institutional, ethical, and scientific responsibility for the complete contents of this paper.

References

Alessandro Achille and Stefano Soatto. Emergence of invariance and disentanglement in deep representations. *Journal of Machine Learning Research*, 19(50):1–34, 2018.

Carter Blair, Kate Larson, and Edith Law. Reflective verbal reward design for pluralistic alignment. In *Proceedings of the 34th International Joint Conference on Artificial Intelligence (IJCAI)*, 2025. doi: 10.24963/ijcai.2025/1141.

Carter Blair, Ben Armstrong, Shiri Alouf-Heffetz, Nimrod Talmon, and Davide Grossi. Probably approximately consensus: On the learning theory of finding common ground. *arXiv preprint arXiv:2604.21811*, 2026a.

Carter Blair, Ariel D. Procaccia, and Milind Tambe. Embeddings for preferences, not semantics. *arXiv preprint arXiv:2605.08360*, 2026b.

Niclas Boehmer, Sara Fish, and Ariel D. Procaccia. Generative social choice: The next generation. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, volume 267 of *Proceedings of Machine Learning Research*, pp. 4627–4659, 2025.

Xingyu Chen, Brandon Fain, Liang Lyu, and Kamesh Munagala. Proportionally fair clustering. In *Proceedings of the 36th International Conference on Machine Learning (ICML)*, volume 97 of *Proceedings of Machine Learning Research*, pp. 1032–1041, 2019.

Michal Feldman, Amos Fiat, and Iddan Golomb. On voting and facility location. In *Proceedings of the 2016 ACM Conference on Economics and Computation (EC '16)*, pp. 269–286, 2016. doi: 10.1145/2940716.2940725.

Sara Fish, Paul Gölz, David C. Parkes, Ariel D. Procaccia, Gili Rusak, Itai Shapira, and Manuel Wüthrich. Generative social choice. *Journal of the ACM*, 73(2), 2026. doi: 10.1145/3799709.

Tianyu Gao, Xingcheng Yao, and Danqi Chen. SimCSE: Simple contrastive learning of sentence embeddings. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 6894–6910, 2021. doi: 10.18653/v1/2021.emnlp-main.552.

Leon Kellerhals and Jannik Peters. Proportional fairness in clustering: A social choice perspective. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 37, pp. 28471–28494, 2024.

Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 33, pp. 9459–9474, 2020.

Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. Lost in the middle: How language models use long contexts. *Transactions of the Association for Computational Linguistics*, 12:157–173, 2024. doi: 10.1162/tacl_a_00638.

Niklas Muennighoff, Nouamane Tazi, Loic Magne, and Nils Reimers. MTEB: Massive text embedding benchmark. In *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics (EACL)*, pp. 2014–2037, 2023. doi: 10.18653/v1/2023.eacl-main.148.

Rodrigo Nogueira and Kyunghyun Cho. Passage re-ranking with BERT. *arXiv preprint arXiv:1901.04085*, 2019.

Nils Reimers and Iryna Gurevych. Sentence-BERT: Sentence embeddings using siamese BERT-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP-IJCNLP)*, pp. 3982–3992, 2019. doi: 10.18653/v1/D19-1410.

Melanie Sclar, Yejin Choi, Yulia Tsvetkov, and Alane Suhr. Quantifying language models' sensitivity to spurious features in prompt design or: How I learned to start worrying about prompt formatting. In *International Conference on Learning Representations (ICLR)*, 2024.

Christopher T. Small, Michael Bjorkegren, Timo Erkkilä, Lynette Shaw, and Colin Megill. Polis: Scaling deliberation by mapping high dimensional opinion spaces. *RECERCA. Revista de Pensament i Anàlisi*, 26(2):1–26, 2021. doi: 10.6035/recerca.5516.

Michael Henry Tessler, Michiel A. Bakker, Daniel Jarrett, Hannah Sheahan, Martin J. Chadwick, Raphael Koster, Georgina Evans, Lucy Campbell-Gillingham, Tantum Collins, David C. Parkes, Matthew Botvinick, and Christopher Summerfield. AI can help humans find common ground in democratic deliberation. *Science*, 386(6719):eadq2852, 2024. doi: 10.1126/science.adq2852.

Vijay Viswanathan, Kiril Gashteovski, Carolin Lawrence, Tongshuang Wu, and Graham Neubig. Large language models enable few-shot clustering. *Transactions of the Association for Computational Linguistics*, 12:321–333, 2024. doi: 10.1162/tacl_a_00648.

Peiyi Wang, Lei Li, Liang Chen, Zefan Cai, Dawei Zhu, Binghuai Lin, Yunbo Cao, Qi Liu, Tianyu Liu, and Zhifang Sui. Large language models are not fair evaluators. *arXiv preprint arXiv:2305.17926*, 2023.

Yuwei Zhang, Zihan Wang, and Jingbo Shang. ClusterLLM: Large language models as a guide for text clustering. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 13903–13920, 2023. doi: 10.18653/v1/2023.emnlp-main.858.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. Judging LLM-as-a-judge with MT-bench and chatbot arena. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, volume 36, 2023.

A Algorithmic Flow of the Placement Cascade

Algorithm: Hierarchical Online Placement Cascade
Input: Statement s, Question Q, Vector Index I, Thresholds $\tau_{\text{attach}}, \tau_{\text{synth}}, \tau_{\text{cluster}}, \tau_{\text{cohesion}}$.

1. $b \leftarrow \textsc{DistillBrief}(s)$
2. $\psi(s) \leftarrow \textsc{EmbedText}(\text{“Question: ”} + Q + \text{“\textbackslash nAnswer: ”} + b)$
3. $C \leftarrow \textsc{AnnQuery}(I, \psi(s), k = 15)$
4. **for each** candidate synthesis $S \in C$ **do:**
   - **if** $\textsc{Evidence}(s, S) \geq \tau_{\text{attach}} \wedge \textsc{CohesionGate}(s, S, \tau_{\text{cohesion}})$ **then:**
     - **if** $\textsc{EquivalenceJudge}(s, S.\text{text}) = \texttt{same}$ **then:**
       - $\textsc{AttachStatementToSynthesis}(s, S)$; **return** ATTACHED
5. **for each** unattached statement $u \in C$ **do:**
   - **if** $\cos(\psi(s), \psi(u)) \geq \tau_{\text{synth}} \wedge \textsc{CoherenceCheck}(s, u)$ **then:**
     - $S_{\text{new}} \leftarrow \textsc{SpawnSynthesis}(\{s, u\})$
     - **if** $\textsc{ThemeJudge}(S_{\text{new}}, \textsc{Themes}(Q)) = \texttt{NONE}$ **then:**
       - $\textsc{createThemeForSynth}(S_{\text{new}})$
     - **return** SPAWNED
6. $T \leftarrow \textsc{ThemeJudge}(s, \textsc{Themes}(Q))$; **if** $T \neq \texttt{NONE}$ **then** $\textsc{FileUnderTheme}(s, T)$; **return** THEMED
7. $\textsc{CreateStandingSingleton}(s)$; **return** SINGLETON

B Prompt Specifications and Shipped Operational Rubrics

P1 Equivalence Judge Rubric.

> “`same`: A and B are paraphrases of essentially the same proposal.
> `related`: same topic but different actions, stances, or magnitudes.
> `different`: about different things; wording similar by coincidence.
> `opposite`: contradictory actions on the same subject (‘ban X’ vs ‘permit X’).
> Use ‘same’ ONLY when the two proposals are essentially interchangeable. When in doubt between ‘same’ and ‘related’, prefer ‘related’.”

Synthesis Writer Rubric.

> “First check: (1) directional conflict (inputs pull opposite ways on the same lever); (2) distinct actions within the same theme (different specific interventions that happen to address related problems). If a supporter could reasonably back one while opposing the other’s intervention, the inputs are distinct ideas. When in doubt, refuse.
> Formulation clause: formulation differences are not grounds for refusal; refuse only when the difference IS the intervention.
> Scope rule: never widen the scope. Never add mechanisms, criteria, or commitments that no input stated.”

C Real-Data Deliberation Audit Protocols

The deliberation collected 114 statements on environmental research translation. Blind merge auditors evaluated all constituent members against the core proposal under the pipeline’s own rubric, with doubt counted as refutation. Missed-merge sweepers evaluated unplaced statements and non-merged pairs. Text fidelity was graded via an automated rubric assessing whether constituent asks were preserved, weakened, or lost, and flagging any unstated commitments. Both replays ran with the Claim Registry disabled, which is its shipped default.